

Phenolic substance characterization and chemical and cell-based antioxidant activities of 11 lentils grown in the northern United States.

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Chemical and cellular antioxidant activities and phenolic profiles of 11 lentil cultivars grown in the cool northern parts of the United States were investigated. Individual phenolic compounds, including phenolic acids, flavan-3-ols, flavones, and anthocyanins, were further quantitatively investigated by HPLC. Cellular antioxidant activities (CAA) and peroxy radical scavenging capacity (PRSC) were evaluated by fluorescence microplate reader. Cultivar Morton exhibited the highest individual flavan-3-ols (catechin and epicatechin) and total flavonoids, as well as the highest antioxidant properties (PRSC and CAA) among all lentils tested. Five phenolic acids of the benzoic types and their derivatives (gallic, protocatechuic, 2,3,4-trihydroxybenzoic, p-hydroxybenzoic acid, and protocatechualdehyde) and four phenolic acids of the cinnamic type (chlorogenic, p-coumaric, m-coumaric, and sinapic acid) were detected in all lentil cultivars. Two flavan-3-ols [(+)-catechin and (-)-epicatechin] and one flavone (luteolin) were detected in all lentil cultivars. Among all phenolic compounds detected, sinapic acid was the predominant phenolic acid, and (+)-catechin and (-)-epicatechin were the predominant flavonoids. These results showed that different phenotype lentils possessed considerable variations in their individual phenolic compounds, as well as chemical and cellular antioxidant activities. Caffeic acid, catechin, epicatechin, and total flavonoids significantly ($p < 0.05$) correlated with peroxy radical scavenging assay. Cellular antioxidant assay significantly correlated with chemical antioxidant assay ORAC. The results from this study could be very interesting for breeding programs to improve lentils for use as functional foods.